Can Chance Explain the Mean Difference?

Unit 4 · Topic 4.2 · THE PROBLEM

Suppose 50 student volunteers test a planning prompt for a logic task. They are randomly assigned to two groups of 25. Mean times are 14.2 minutes with the prompt and 17.8 with standard directions: prompt minus standard is -3.6. Under no effect, a simulation shuffles the same times into new groups. Only 18 of 2,000 shuffled differences are -3.6 or lower.

Nia: "This is evidence that the prompt lowered mean time for these volunteers, but we could still be wrong."

Owen: "The 0.009 is the probability of no effect, so there is a 99.1% chance the prompt works for every district student."

Experiment results

Group	n	Mean (min)
Prompt	25	14.2
Standard	25	17.8

FIRST TAKE — one sentence before you work the parts

Whose claim sounds more believable, Nia's or Owen's? Give one reason from the study.

DOK 1 **(a)** What difference in group means would we expect if the prompt had no effect?

DOK 2 **(b)** Compute 18/2,000. In one sentence, explain what this proportion estimates if the prompt has no effect.

DOK 3 **(c)** In 3–4 sentences, decide whose claim is better supported. Use the simulation result, explain why an error is still possible, and say why the volunteers do not represent every district student.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.